PADHAKU
Science · Class 10 Science



CHAPTER 6

Control and Coordination

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.

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⭐ In-text · 6.1 Animals – Nervous System

? **Look at Fig. 6.1 (a) and identify the parts of a neuron (i) where information is acquired, (ii) through which information travels as an electrical impulse, and (iii) where this impulse must be converted into a chemical signal for onward transmission.**

✔ ANSWER

(i) **Dendrites** (the dendritic tips) acquire the information.

(ii) The impulse travels from the dendrite to the **cell body** and then along the **axon**.

(iii) At the **end of the axon (nerve ending)** the electrical impulse is converted into chemicals that cross the synapse.

🧪 Activity 6.1

1 **Put some sugar in your mouth, then block your nose and eat sugar again; also block your nose while eating lunch. Is there a difference in how sugar and food taste if your nose is blocked? If so, why? Do you come across a similar situation when you have a cold?**

✔ ANSWER

Observation: With the nose blocked, sugar still tastes sweet but the food tastes dull; its full flavour is not felt.

Reason: Flavour is detected by both gustatory receptors (taste, tongue) and olfactory receptors (smell, nose). With the nose blocked, smell does not reach the olfactory receptors, so the brain gets less information.

Yes: during a cold the nose is blocked, so food seems tasteless.

Conclusion: Taste and smell together decide how we sense the flavour of food.

⭐ In-text · 6.1.1 What happens in Reflex Actions?

? Can you now trace the sequence of events which occur when a bright light is focussed on your eyes?

✔ ANSWER

- Receptors in the eye (retina) detect the bright light (stimulus).
- A sensory neuron carries the impulse to the brain.
- The impulse is passed on through a relay neuron to a motor neuron.
- The motor neuron carries the impulse to the muscles of the iris (effector).
- The iris muscles act and the **pupil becomes smaller**, so less light enters the eye (we may also blink).

This happens automatically, without thinking: a reflex action.

⭐ In-text · 6.1.2 Human Brain

? Study the labelled diagram of the human brain (Fig. 6.3). Can we find out the function of each part?

💡 SAMPLE ANSWER

Part	Function
Fore-brain (cerebrum)	Main thinking part; receives sensory impulses (sight, hearing, smell...); association areas; motor areas control voluntary muscles
Hypothalamus (fore-brain)	Hunger/fullness and body temperature; controls release of many hormones from the pituitary
Mid-brain	Controls some involuntary actions and reflexes, e.g. change in size of the pupil
Pons (hind-brain)	Helps regulate breathing; relays messages
Medulla (hind-brain)	Involuntary actions: blood pressure, heartbeat, salivation, vomiting
Cerebellum (hind-brain)	Precision of voluntary actions; posture and balance
Pituitary gland	Secretes hormones such as growth hormone

Spinal cord	Reflex actions; carries messages between brain and body

⭐ In-text · 6.1.4 How does the Nervous Tissue cause Action?

? Based on what we have discussed, what do you think the differences between voluntary and involuntary muscles would be?

💡 SAMPLE ANSWER

- **Voluntary muscles** (e.g. leg, arm muscles) move when we decide; they get impulses from the motor areas of the fore-brain.
- **Involuntary muscles** (e.g. heart, stomach, intestine, blood vessels) move on their own, without our thinking; they are controlled by the mid-brain and hind-brain (medulla).

In both, special proteins in the muscle cells change shape and arrangement to make the cells shorter.

❓ Questions · 6.1 Animals – Nervous System

Q 1 What is the difference between a reflex action and walking?

✔ ANSWER

Reflex action	Walking

Sudden, automatic response to a stimulus, without thinking	Voluntary action, done by our will, after thinking
Controlled mainly by the spinal cord (reflex arc)	Controlled by the fore-brain; precision and balance by the cerebellum
Very quick; we cannot control it	Slower; we can start, stop or change it

Q 2 **What happens at the synapse between two neurons?**

✔ ANSWER

The electrical impulse reaching the end of the axon releases some **chemicals**.

These chemicals cross the gap (synapse) and start a similar **electrical impulse in a dendrite of the next neuron**.

So the message passes in one direction from one neuron to the next.

Q 3 **Which part of the brain maintains posture and equilibrium of the body?**

✔ ANSWER

The **cerebellum** (part of the hind-brain).

Q 4 **How do we detect the smell of an agarbatti (incense stick)?**

✔ ANSWER

Smell particles from the agarbatti spread in the air and enter our nose.

The **olfactory receptors** in the nose detect them and create electrical impulses.

Sensory nerves carry these impulses to the smell area of the **fore-brain**, which interprets them as the smell of agarbatti.

Q 5 **What is the role of the brain in reflex action?**

✔ ANSWER

Reflex arcs are formed in the **spinal cord**, so the quick response happens without the brain thinking about it.

But the information input also goes on to reach the **brain**, so the brain becomes aware of what happened (e.g. we feel pain) and can plan further actions.

Some reflexes, such as change in the size of the pupil, are controlled by the brain (mid-brain).

Activity 6.2

1 **Germinated bean seeds are kept on a wire mesh over a flask of water in a box open on one side facing light. After 2–3 days the shoots bend towards light and roots away from light. The flask is then turned so that**

shoots are away from light and roots towards light.

Have the old parts of the shoot and root changed direction? Are there differences in the direction of the new growth? What can we conclude?

✔ ANSWER

Old parts: No, the old parts of the shoot and root do not change direction.

New growth: Yes. The new part of the shoot again bends towards light and the new part of the root bends away from light.

Conclusion: Shoots are positively phototropic and roots are negatively phototropic. The bending is due to growth in a particular direction (tropic movement), so only growing parts respond.

⭐ In-text · 6.2.2 Movement Due to Growth

? **Shoots bend towards light while roots bend away from it. How does this help the plant?**

✔ ANSWER

Shoots bending towards light get more sunlight for **photosynthesis**.

Roots growing away from light (and down into the soil) **fix the plant** firmly and absorb **water and minerals**.

? **If ‘hydro’ means water and ‘chemo’ refers to chemicals, what would ‘hydrotropism’ and ‘chemotropism’ mean? Can we think of examples?**

✔ ANSWER

Hydrotropism: directional growth of a plant part in response to water. Example: roots grow towards moist soil / a water source.

Chemotropism: directional growth in response to chemicals. Example: growth of the pollen tube towards the ovule.

❓ Questions · 6.2 Coordination in Plants

Q 1 **What are plant hormones?**

✔ ANSWER

Plant hormones (phytohormones) are **chemical substances** made in plants that help to coordinate **growth, development and responses to the environment**.

They are made at places away from where they act and diffuse to the area of action.

Examples: auxins, gibberellins, cytokinins, abscisic acid.

Q 2 **How is the movement of leaves of the sensitive plant different from the movement of a shoot towards light?**

✔ ANSWER

Leaves of sensitive plant	Shoot towards light
Response to touch	Response to light (phototropism)
Not due to growth	Due to growth (directional)
Very quick	Slow
Cells change shape by changing their water content	Auxin makes cells on the shady side grow longer
Not directional	Directional (towards the stimulus)

Q 3 **Give an example of a plant hormone that promotes growth.**

✔ ANSWER

Auxin (also gibberellins, which help stem growth, and cytokinins, which promote cell division).

Q 4 **How do auxins promote the growth of a tendril around a support?**

✔ ANSWER

When a tendril touches a support, auxin moves to the side **away from the support**.

Cells on that side grow longer faster, while the side in contact with the support grows more slowly.

This unequal growth makes the tendril **coil around the support** and cling to it.

Q 5 **Design an experiment to demonstrate hydrotropism.**

💡 **SAMPLE ANSWER**

- Take a box (or trough) filled with dry soil and sow a germinating seed (e.g. bean) in it.
- Bury a small porous clay pot filled with water in the soil a little away from the seed, on one side.
- Water only the clay pot; water slowly seeps out on that side, so that side of the soil stays moist.
- After a few days, gently remove the soil and observe the root.

Observation: The root bends and grows towards the clay pot (moist soil).

Conclusion: Roots grow towards water: they show **positive hydrotropism**.

🧪 Activity 6.3

1 **Look at Fig. 6.7. Identify the endocrine glands mentioned in the figure, and find out about other glands.**

💡 **SAMPLE ANSWER**

Glands shown: **hypothalamus, pituitary gland, pineal gland, thyroid gland, parathyroid glands, thymus, adrenal glands, pancreas, testes** (in males) and **ovaries** (in females).

Other glands (beyond Table 6.1), e.g.:

- Pineal gland: secretes melatonin (sleep–wake cycle).
- Parathyroid: parathormone, controls calcium level in blood.
- Thymus: thymosin, helps immunity in children.

⭐ In-text · 6.3 Hormones in Animals

? **A symptom of goitre is a swollen neck. Can you correlate this with the position of the thyroid gland in Fig. 6.7?**

✔ **ANSWER**

The **thyroid gland is in the neck** (in front of the windpipe).

When iodine is lacking, the thyroid cannot make enough thyroxin and gets enlarged, so the neck looks swollen.

Activity 6.4

1 Complete Table 6.1: Some important hormones and their functions.

S.No.	Hormone	Endocrine gland	Functions
1	Growth hormone	Pituitary gland	Stimulates growth in all organs
2	?	Thyroid gland	Regulates metabolism for body growth
3	Insulin	?	Regulates blood sugar level
4	Testosterone	Testes	?
5	?	Ovaries	Development of female sex organs, regulates menstrual cycle, etc.
6	Adrenaline	Adrenal gland	?
7	Releasing hormones	?	Stimulates pituitary gland to release hormones

✔ ANSWER

S.No.	Hormone	Endocrine gland	Functions

1	Growth hormone	Pituitary gland	Stimulates growth in all organs
2	**Thyroxin**	Thyroid gland	Regulates metabolism for body growth
3	Insulin	**Pancreas**	Regulates blood sugar level
4	Testosterone	Testes	**Development of male sex organs and changes at puberty in males (e.g. deeper voice, beard)**
5	**Oestrogen**	Ovaries	Development of female sex organs, regulates menstrual cycle, etc.
6	Adrenaline	Adrenal gland	**Prepares the body for emergency: faster heartbeat and breathing, more blood to skeletal muscles**

7	Releasing hormones	**Hypothalamus**	Stimulates pituitary gland to release hormones

❓ Questions · 6.3 Hormones in Animals

Q 1 **How does chemical coordination take place in animals?**

✔ ANSWER

Chemical coordination takes place through **hormones** secreted by **endocrine glands**.

Hormones are released **directly into the blood** and carried to target organs or tissues, where they act.

Their timing and amount are controlled by **feedback mechanisms** (e.g. more insulin when blood sugar rises).

Q 2 **Why is the use of iodised salt advisable?**

✔ ANSWER

Iodine is needed by the **thyroid gland** to make **thyroxin**, which regulates carbohydrate, protein and fat metabolism for proper growth.

If iodine is lacking in the diet, we may suffer from **goitre** (swollen neck). Iodised salt supplies this iodine.

Q 3 **How does our body respond when adrenaline is secreted into the blood?**

✔ ANSWER

- The **heart beats faster**, supplying more oxygen to the muscles.
- Blood supply to the digestive system and skin is **reduced** (small arteries there contract), and blood is diverted to the **skeletal muscles**.
- The **breathing rate increases** due to contraction of the diaphragm and rib muscles.

Together these prepare the body to fight or run away in an emergency.

Q 4 **Why are some patients of diabetes treated by giving injections of insulin?**

✔ ANSWER

Insulin, made by the **pancreas**, regulates the blood sugar level.

In some diabetic patients the pancreas does not secrete enough insulin, so blood sugar rises and causes harmful effects.

Insulin injections supply the missing hormone and **bring the blood sugar level down** to normal.

Exercises

1 **Which of the following is a plant hormone?**
(a) Insulin (b) Thyroxin (c) Oestrogen (d) Cytokinin

✔ ANSWER

(d) Cytokinin. It promotes cell division in plants; the others are animal hormones.

2 **The gap between two neurons is called a**
(a) dendrite (b) synapse (c) axon (d) impulse

✔ ANSWER

(b) synapse. Chemicals released at the axon end cross this gap to the next neuron.

3 **The brain is responsible for**
(a) thinking (b) regulating the heart beat (c) balancing the body (d) all of the above

✔ ANSWER

(d) all of the above. Fore-brain: thinking; medulla: heartbeat; cerebellum: balance.

4 **What is the function of receptors in our body? Think of situations where receptors do not work properly. What problems are likely to arise?**

✔ ANSWER

Function: Receptors are specialised tips of nerve cells, mostly in sense organs. They **detect stimuli** (light, sound, smell, taste, touch, heat) and start nerve impulses that carry the information to the brain or spinal cord.

If they do not work properly:

- Olfactory receptors blocked (cold): we cannot smell, and food tastes dull.
- Touch/heat receptors damaged: we may not feel a hot object and get burnt or hurt.
- Receptors of eye or ear damaged: problems in seeing or hearing.

So the body cannot respond correctly or quickly to changes and may get harmed.

5 **Draw the structure of a neuron and explain its function.**

✔ ANSWER

Draw the labelled diagram as in Fig. 6.1 (a), with these parts: **dendrites, cell body (with nucleus), axon, nerve ending**.

- **Dendrites:** acquire information from receptors or other neurons; a chemical reaction creates an electrical impulse.
- **Cell body:** receives the impulse from the dendrites and passes it on.
- **Axon:** carries the electrical impulse away from the cell body to its end.

- **Nerve ending:** releases chemicals that cross the synapse to the next neuron, muscle or gland.

Function of a neuron: to receive information and conduct it as electrical impulses from one part of the body to another.

6 **How does phototropism occur in plants?**

✔ ANSWER

Phototropism is the bending of a plant part in response to light.

- The hormone **auxin** is made at the shoot tip.
- When light comes from one side, auxin **diffuses towards the shady side** of the shoot.
- More auxin there makes cells on the shady side **grow longer**.
- So the shoot **bends towards light**. Roots bend away from light.

7 **Which signals will get disrupted in case of a spinal cord injury?**

✔ ANSWER

- **Reflex actions** whose reflex arcs pass through the injured part of the spinal cord.
- **Sensory signals** from body parts below the injury to the brain (we may not feel touch, pain or heat there).

- **Motor signals** from the brain to muscles below the injury, so voluntary movement of those parts is lost.

8 **How does chemical coordination occur in plants?**

✔ ANSWER

Plants have no nervous system; they coordinate through **plant hormones**.

Hormones are made at places away from where they act and **diffuse** to the area of action.

- Auxin: cell elongation, phototropism.
- Gibberellins: growth of the stem.
- Cytokinins: cell division (fruits, seeds).
- Abscisic acid: inhibits growth; causes wilting of leaves.

Thus they coordinate growth, development and responses to the environment.

9 **What is the need for a system of control and coordination in an organism?**

✔ ANSWER

- To **detect changes** in the environment and respond with the **correct movement** at the right time.
- To protect the body from danger (e.g. pulling the hand back from a hot object).

- To make different organs and tissues **work together** (e.g. heart, lungs and muscles while running).
- To control growth, development and body functions (blood sugar, metabolism) and keep a balance inside the body.

10 **How are involuntary actions and reflex actions different from each other?**

✔ ANSWER

Involuntary actions	Reflex actions
Actions of muscles going on all the time, without our thinking	Sudden, quick response to a particular stimulus
Controlled by the mid-brain and hind-brain (medulla)	Controlled mainly by the spinal cord (reflex arc)
Need no outside stimulus; e.g. heartbeat, breathing, digestion, blood pressure	Triggered by a stimulus; e.g. pulling hand from a flame, blinking at bright light
Usually slow and continuous	Very fast, protective

11 **Compare and contrast nervous and hormonal mechanisms for control and coordination in animals.**

✔ ANSWER

Nervous mechanism	Hormonal mechanism
Messages are electrical impulses	Messages are chemicals (hormones)
Carried by nerve cells (neurons)	Carried by blood
Very fast	Slower
Reaches only cells connected by nerves	Can reach all cells of the body
Effect is short-lived; cells need time to reset before the next impulse	Effect is long-lasting and steady

Similarity: both control and coordinate body activities and respond to stimuli; they work together (e.g. hypothalamus links the two).

12 **What is the difference between the manner in which movement takes place in a sensitive plant and the movement in our legs?**

✔ ANSWER

Sensitive plant	Our legs
No nervous or muscle tissue	Nerves and muscles are involved
Information passes from cell to cell by electrical-chemical means, with no special tissue	Information passes as electrical impulses through neurons

Cells change shape by changing the amount of water in them (swelling/shrinking)	Muscle cells change shape because special proteins change their shape and arrangement
Involuntary response to touch	Voluntary movement, controlled by the brain

🎉 **Chapter 6 done! Revise the answers once more tomorrow.**